

6	(a) (i) force = mg along the direction of the field/of the motion	M1	A1	[2]
	(ii) no force	B1		[1]
	(b) (i) force due to E -field downwards so force due to B -field upwards into the plane of the paper	B1	B1	[2]
	(ii) force due to magnetic field = Bqv force due to electric field = Eq (<i>use of F_B and F_E not explained, allow 1/2</i>)	B1	B1	
	forces are equal (and opposite) so $Bv = E$ or $Eq = Bqv$ so $E = Bv$	B1		[3]
	(c) sketch: smooth curved path in 'upward' direction	M1	A1	[2]